3 – Swarm Robotics

Swarm Intelligence, Flocking, PSO, ACO

Swarm Robotics: Introduction

- Some animal societies display coordinated and purposeful navigation of several individuals (without a predefined leader)
  - E.g. flocking, schooling, etc.

- Each individual uses only local information about the presence of other individuals and of the environment

- Sometimes there is a predefined leader, but individuals still only use local information to decide how to move
  - E.g. herding, v-formations, processions, etc.

Swarm Robotics: Swarm Intelligence

- **Swarm Intelligence:** emergent collective intelligence of groups of (simple) individuals

1. The swarm can solve complex problems that a single individual could not solve
2. The swarm is composed of several individuals, some of which may be lost or make mistakes, but the overall performance is not affected by this
3. Individuals in the swarm have local sensory information, perform simple action, and have little to no memory; they do not know the global state of the swarm (nor its goal)

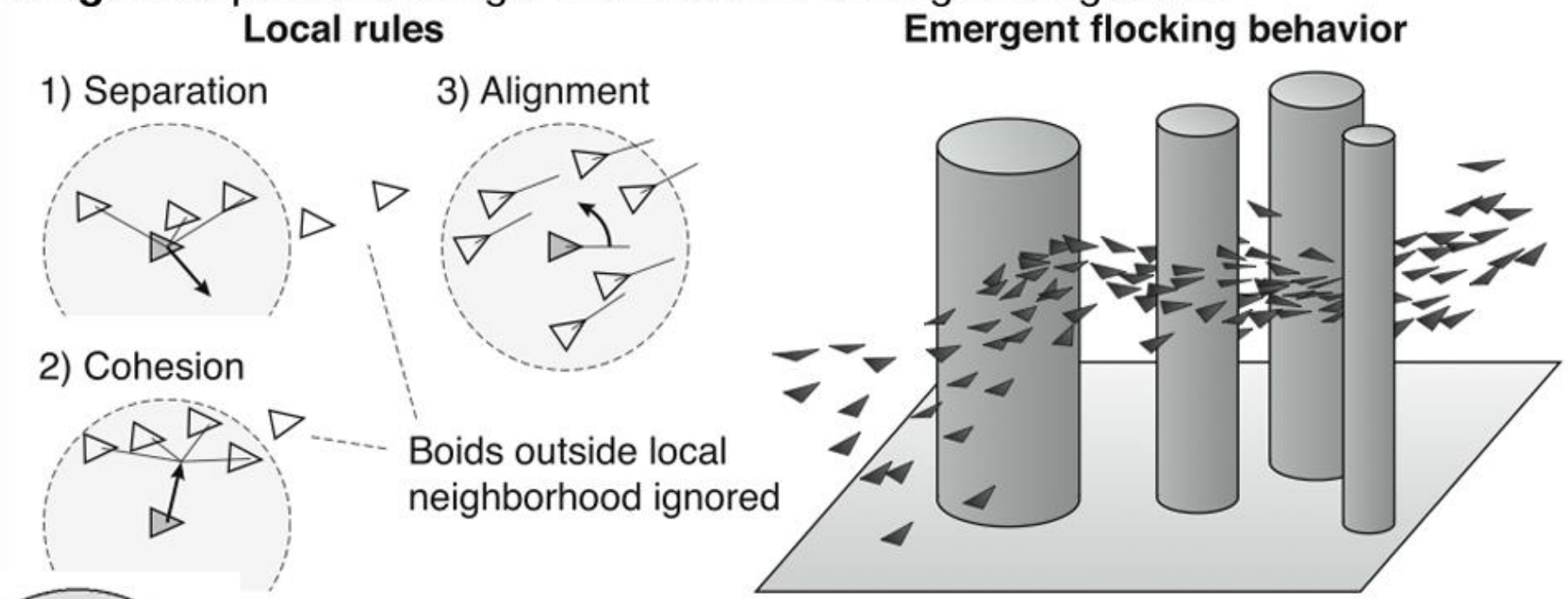
Swarm Robotics: Swarm Intelligence

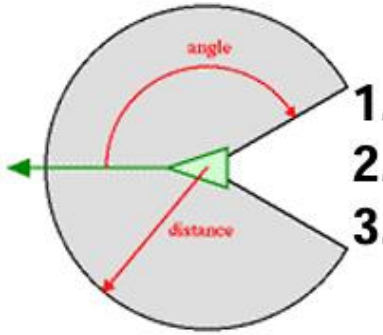
- Swarm intelligence challenges:
  - Difficult to predict emergent behaviour for given individual behaviour
  - Difficult to find individual behavioural rules that result in desired swarm behaviour (reverse engineering)
  - Difficult to ensure that the emerged behaviour is stable

- Say you want to move a series of Autonomous Guided Vehicles (AGVs) through a landscape with narrow passages
- How to coordinate this behaviour without pre-planning or global navigation?

Swarm Robotics: Reynolds Flocking

Sensing: Boid perceives angle and distance of neighboring boids



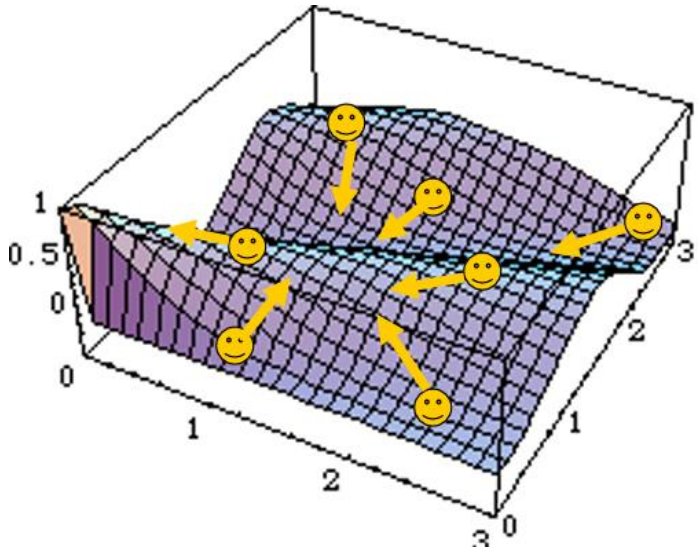


1. **Separation:** Boid maintains a given distance from other boids
2. **Cohesion:** Boid moves towards center of mass of neighboring boids
3. **Alignment:** Boid aligns its angle along those of neighboring boids

Swarm Robotics: PSO

- **Particle Swarm Optimization** (PSO) is an optimization algorithm that can find an optimal value by exploring a fitness landscape
- Imagine a flock of birds that want to find the area with the highest concentration of food. The birds do not know where that area is, but each bird can shout to its neighbours how much food there is at their location
- Most likely to succeed when the birds combine 3 strategies:
  - **Brave:** keep flying in the same direction
  - **Conservative:** fly back towards its own best previous position
  - **Swarm:** fly towards its best neighbour

Swarm Robotics: PSO



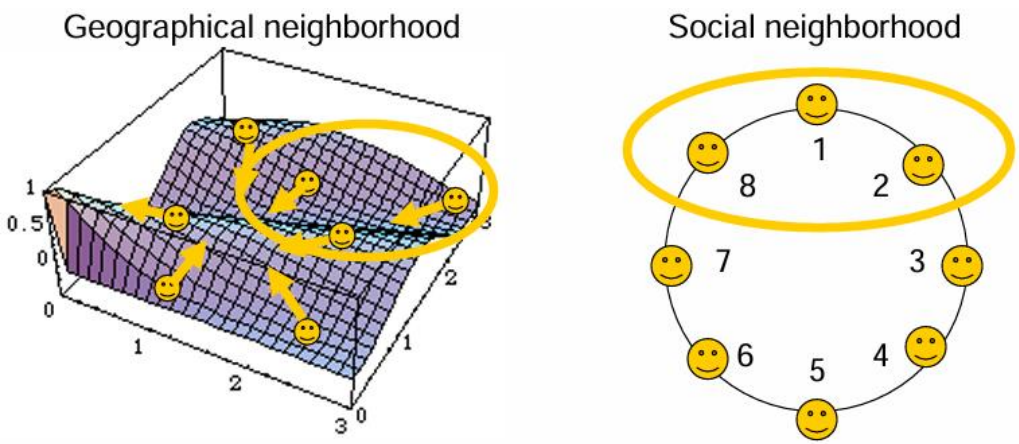
- Each particle *p* is described by:
  - Its **position** *s* (e.g. x and y)
  - Its **velocity** *v* (e.g. the angle and distance of next step)
  - Its **performance** *f* (e.g. the value of the function at its location)

- A particle perceives performances and positions of neighbouring particles, and keeps track of the best position it found so far
- A particle computes the next position by considering its current velocity, the direction to its previous best position, and the direction to its best neighbour (with some error)

$$\boldsymbol{v}_p(t+1) = a \cdot \boldsymbol{v}_p(t) + b \cdot R \cdot \left(\boldsymbol{s}_{pbest} - \boldsymbol{s}_p(t)\right) + c \cdot R \cdot \left(\boldsymbol{s}_{gbest} - \boldsymbol{s}_p(t)\right)$$

Swarm Robotics: PSO

- PSO requires a low number of computations, and can easily be executed directly on the robot if required
- It does not require gradients, and still performs well on more complex search spaces
- Most problems are solved with around 20 particles initialized at random positions, with a **neighbourhood size** of around 3-5



Swarm Robotics: PSO vs Artificial Evolution

- Like artificial evolution, PSO works with a population and some random factor to update solutions

- Unlike artificial evolution, there is no generation change, genome, or competition between solutions (rather cooperation)

- The biggest challenge for PSO is the transformation of the problem so that it can be encoded and search by particles

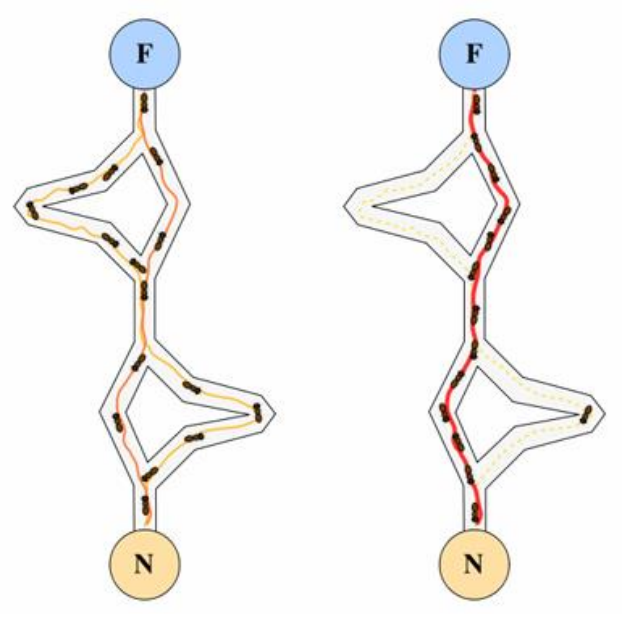
Swarm Robotics: Ant Colony Optimization

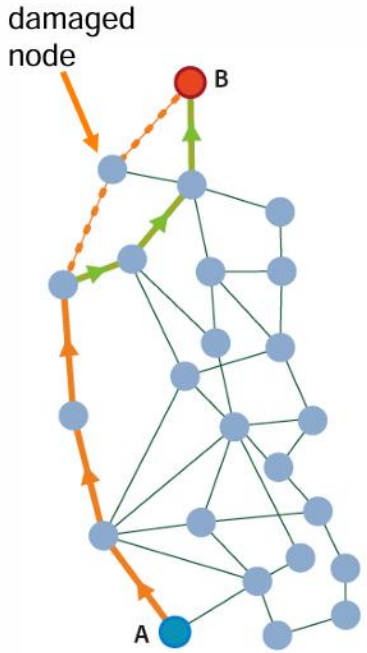
- **Ant Colony Optimization** (ACO) is an algorithm designed to solve TSP-like problems where a shortest path needs to be found

- Ants leave a **pheromone** trail behind as they walk, allowing other ants to follow their path (this trail decays over time)
- Ants follow the path with the highest pheromone concentration, or choose randomly if there is no pheromone
- Since the shortest path has a higher throughput, and therefore will have a higher pheromone concentration

Swarm Robotics: Ant Colony Optimization

- A huge advantage of ACO is that it allows for **dynamic rerouting**
- If a node suddenly breaks, the ants will find a new shortest path





Swarm Robotics: Ant Colony Optimization

- ACO finds the optimal solution up to around 30 cities, and still finds good solutions on larger problems (comparatively)
- Usually around 100 ants, distributed over random nodes
- The pheromone levels of edges are updated based on the length of the edge, the current amount of pheromone, and evaporation

Swarm Robotics: Examples

Werfel et al. (THERMES)	**Rubenstein et al.** (Kilobots)
Assembly though manipulation of environment, robot swarm builds final structure from bricks	Self-assembly, Robots form/are part of final structure
Only coordination through environment (check what has already been built)	Direct local communication between robots, message passing to count number of hops to infer long-distance knowledge
A few robots building a 3D structure	More than 1000 robots forming a 2D structure
Decentralized, all robots run same code, all robots know final structure, all robots make decisions based on local sensor readings Even for relatively simple systems very slow!	